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Binding of per- and polyfluoroalkyl substances with liver and serum proteins in rats: implications for physiologically based pharmacokinetic modelling

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ABSTRACT

Understanding the binding between per- and polyfluoroalkyl substances (PFAS) and proteins is essential for elucidating their toxicokinetics and tissue distribution. Here, we quantified the binding affinities of 14 PFAS to rat liver fatty acid-binding protein (rL-FABP) and rat serum albumin (RSA). Results showed that PFAS exhibit strong binding affinities (K_a) to the rL-FABP ($10^3 \sim 10^5$ M^{-1}), particularly among medium- to long-chain perfluorinated carboxylic acids (PFCAs). The binding affinity of PFAS to RSA ranged from approximately 10^4 to 10^5 M^{-1}, with 1 to 4 binding sites. Molecular docking results supported that PFAS binding to proteins is an exothermic process driven by van der Waals forces, hydrogen bonding, and electrostatic interactions. Additionally, long-chain PFCAs were shown to adopt a "U"-shaped conformation within the ligand-binding cavities of rL-FABP and RSA. The newly developed physiologically based pharmacokinetic model using measured binding data demonstrates a substantial improvement in the goodness of fit to experimental observations, reducing the prediction error by 20 %~216 %. Finally, we found that the PFAS liver-blood partition could be mainly explained by the binding affinity ratios of PFAS to liver and blood proteins, which could be further extrapolated from rats to humans, providing useful insights to understand the tissue distribution of PFAS.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are synthetic organic compounds that vary from one fully fluorinated carbon to three connected carbons, each with two or more fluorine bonds, known for their persistence, bioaccumulation and toxicity (Evich et al., 2022; Hammel et al., 2022; Hopkins et al., 2023; Li et al., 2024). In biological systems, their amphiphilic nature drives associated toxicokinetics (such as absorption, distribution) (Jia et al., 2022; Smeltz et al., 2023; Vujic et al., 2024), and both free and protein-bound PFAS are widely distributed throughout the body via systemic circulation (Ryu et al., 2024). In particular, these compounds are able to cross critical biological barriers,

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including the placental and blood–brain barriers, and to compete with endogenous molecules within tissues (Ebert et al., 2020). Understanding the mechanisms of PFAS toxicokinetics is essential for assessing their toxicity and health implications.

The toxicokinetics of PFAS in biological systems is governed by a complex array of cellular and molecular mechanisms, including protein binding, polar lipids, transmembrane transport, nuclear receptor interactions, and hormonal regulation (Chen et al., 2023; Søderstrøm et al., 2022; Zhao et al., 2023). Of these, protein binding is one of the important processes (Fischer et al., 2024; Jia et al. 2022; Ryu et al. 2024; Zhao et al. 2023), with binding proteins such as liver fatty acid-binding protein (L-FABP) and serum albumin playing integral roles in the transport and storage of PFAS within the body (Zhao et al. 2023). L-FABP, highly expressed in the liver, has a strong affinity for long-chain fatty acids, making it a significant determinant of PFAS accumulation in hepatic tissue, due to the structural similarities between PFAS and fatty acids (Han et al., 2021; Zhang et al., 2013). Meanwhile, serum albumin, the predominant protein in plasma in mammals, modulates the distribution and transport of PFAS in the bloodstream through its binding (Fischer et al. 2024). Proteins in the liver and blood have been shown to compete for binding with PFAS compounds, potentially altering their toxicokinetics and tissue distribution, as well as toxicity (Fischer et al. 2024; Jia et al. 2022). While previous research that has reported the PFAS binding with proteins has predominantly focused on human proteins, only two studies concerning PFAS binding to rat proteins (Han et al., 2003; Woodcroft et al., 2010), limiting our understanding in the toxicokinetics across species. Among the two studies, one study has quantified the binding affinity of rat L-FABP (rL-FABP) for a couple of perfluoroalkyl carboxylic acids (PFCAs), with a particular focus on short-chain PFCAs (Woodcroft et al. 2010). Regarding rat serum albumin (RSA), there is only one study that measured the binding affinity of perfluorooctanoic acid (PFOA) (Han et al. 2003). The shortage in the systematic examinations on PFAS binding to protein in rats has constrained further toxicokinetic analyses, as well as constrains the extrapolation of specific toxicological mechanisms, particularly those relevant to human populations (Ryu et al. 2024).

In addition, the liver plays a central role in the toxicokinetics of PFAS due to its direct connection to systemic circulation. Understanding the accumulation of PFAS in the liver is critical for elucidating their behavior in biological systems (Vujic et al. 2024; Zhu et al., 2023b). The liver-blood partition of PFAS is one of the important parameters, reflecting the equilibrium of pollutant distribution between the liver and blood (Baumert et al., 2023; Nielsen et al., 2024). While previous studies have mapped PFAS concentrations across human organs through post-mortem analyses(Baumert et al. 2023; Pérez et al., 2013), these measurements primarily reflect terminal accumulation states rather than dynamic partitioning mechanisms. Physiologically, the liver-blood partition coefficient of individual PFAS is influenced by the compound's chemical properties, its binding affinity to liver and blood proteins, liver metabolic activity, and other factors (Baumert et al. 2023; Jia et al. 2022; Vujic et al. 2024). However, the mechanisms governing the liver-blood partitioning of PFAS remain poorly characterized. Given the strong affinity of PFAS for liver and blood proteins, protein binding affinity could be the critical influencing factor for determining liver-blood partition coefficients for PFAS. Crucially, elucidating protein-driven liver-blood partitioning dynamics allows interspecies extrapolation of PFAS distribution patterns, generating vital toxicokinetic parameters for human health risk assessment. To the best of our knowledge, only Jia et al. (2022) has explored a negative correlation between the PFAS liver-blood partition coefficient in mice and the dissociation constants of human proteins. Nonetheless, such association remains unclear for other species, for example rats studied here.

Given the strong binding affinities of PFAS with RSA and L-FABP, we hypothesize that protein-mediated interactions drive organ-specific accumulation patterns and interspecies toxicokinetic variations. However, systematic investigations into PFAS-protein binding mechanisms in rats remain critically underexplored, particularly binding stoichiometry, and their quantitative linkages to tissue-specific partitioning. To investigate whether the binding of PFAS to proteins influences their pharmacokinetics, organ distribution patterns, and interspecies variations in biological systems, we conducted the following exploration: firstly, this study investigated the binding affinities of 14 PFAS to rL-FABP and RSA through a series of *in vitro* experiments and computational modeling. Secondly, the results were then utilized to develop a physiologically based pharmacokinetic (PBPK) model of PFAS for male rats to understand the toxicokinetics of PFAS. In addition, we explored the association between protein binding of PFAS and liver-blood partition. These findings may highlight the critical role of rL-FABP and RSA in modulating the toxicokinetics and tissue distribution of PFAS, offering some new insights in understanding the bioaccumulation of PFAS in biological systems.

2. Materials and methods

This study employed integrated experimental and computational approaches to determine the binding affinities and stoichiometry of PFAS with rat liver fatty acid-binding protein (rL-FABP) and rat serum albumin (RSA), refined a PBPK model incorporating protein–ligand binding and active transport mechanisms, and analyzed the mechanistic association between PFAS liver-to-blood partition coefficients and protein interaction dynamics (Fig. S1).

2.1. Quantification of PFAS-Protein binding affinity constants

The binding constants of PFAS to rL-FABP were quantified via fluorescence displacement assays, while the interaction parameters between PFAS and RSA were determined using integrated ultrafiltration and isothermal titration calorimetry (ITC) techniques. Molecular docking simulations were further employed to validate the experimentally identified binding sites and energetically characterize the ligand–protein interaction affinities.

2.1.1. Chemicals and proteins

The PFAS selected in this study included ten perfluoroalkyl carboxylic acids (PFCAs) with chain lengths ranging from C4 to C14, three perfluoroalkyl sulfonic acids (PFSAs) with chain lengths ranging from C4 to C8, and one alternative perfluorinated compound (GenX). Details regarding the chemical names are provided in Supplementary Materials (SM Table S1). Internal standards and native PFAS were purchased from Wellington Laboratories (Ontario, Canada), and 1-anilinonaphthalene-8-sulfonic acid (1,8-ANS) was acquired from Thermo Fisher Scientific. RSA free of saturated fatty acids, was purified through the centrifugation of rat serum, which was purchased from Cusabio Technology LLC. rL-FABP was expressed in prokaryotic systems and obtained from Novoprotein Scientific (Shanghai) Inc. The SDS-PAGE results of rL-FABP and RSA are shown in Fig. S2.

2.1.2. Binding affinity of PFAS to rL-FABP

L-FABP is a small molecular protein with a single binding cavity, where fluorescent probes bind to the active site, exhibiting fluorescence properties (Yang et al., 2020; Zhang et al. 2013). This study employed an efficient fluorescence displacement method to investigate the binding affinity of PFAS to recombinant rL-FABP, refer to the method of Yang et al. (2020). The binding affinity of PFAS to rL-FABP was evaluated by assessing the displacement of the fluorescent probe 1,8-ANS from the 1,8-ANS-rL-FABP complex. The experiment was conducted in two parts: direct ligand measurement and competitive ligand measurement.

For the direct ligand measurement, 1 μL of 1,8-ANS at varying concentrations was injected into 100 μL of the rL-FABP buffer system (1 μM). Fluorescence changes were monitored at an excitation wavelength of 350 nm and an emission range of 420–600 nm to determine the number of binding sites, saturation concentration, and binding affinity

of 1,8-ANS to rL-FABP. In the competitive ligand measurement, 1 μL of PFAS solution at different concentrations was added to 100 μL of the 1,8-ANS-rL-FABP complex (with 1,8-ANS at saturation). The decrease in fluorescence at the same excitation and emission wavelengths were recorded to calculate the binding affinity of PFAS to rL-FABP. Oleic acid (OA) served as a positive control to validate the method's feasibility, and all experiments were conducted in triplicate. Detailed experimental methods are provided in the SM Text S1.

2.1.3. Binding affinity of PFAS to RSA

Given the larger molecular size of RSA and its multiple drug-binding sites, ITC (Wu et al., 2009) and ultrafiltration (Beesoon and Martin 2015) were utilized in this study to determine the binding constants of PFAS with RSA. In the ITC experiment, the PFAS solution was incrementally injected into the protein solution at a controlled rate and temperature, allowing for the measurement of thermodynamic parameters such as enthalpy changes. The use of Nano-ITC (TA Instruments) increased the sensitivity of heat change detection. Both RSA and PFAS were dissolved in the same buffer containing 1 % (v/v) Dimethyl sulfoxide (DMSO) to improve solubility, and all working solutions were properly degassed. The sample cell was loaded with RSA (0.02 mM, effective volume 170 μL), while the injection syringe contained the PFAS solution (0.6 mM, effective volume 50 μL). The injection volume is set to 1.96 μL, with a 150-second interval between injections. Titration was performed at 37 °C and 350 rpm to ensure equilibrium at each titration point, involving 25 independent injections per experiment. The initial injection volume (1.96 μL) was excluded from the analysis. All experiments were conducted in triplicate. Prior to the experiments, pure water was titrated against itself to verify the stability of the instrument and methodology. The heat of ligand dilution was subtracted in all experiments. Data analysis was conducted using Nano-analyze software, fitting the integrated heat signal to the Independent model. The dissociation constant (K_d, $K_a = K_d^{-1}$), enthalpy change (ΔH), and the number of binding sites (n) were determined through curve fitting (Lima Cavalcanti et al., 2023; Moro et al., 2022; Woodcroft et al. 2010).

The ultrafiltration method was adapted based on improvements reported by Gao et al., (2019) and Sheng et al., (2016). Briefly, 1 μL of PFAS stock solution (0.2–15 mM) was added to 100 μL of RSA solution (1 μM, dissolved in 0.1 M PBS, pH 7.4) and incubated for 1 h in a 37 °C circulating water bath. Subsequently, 700 μL of the mixed solution was transferred to an ultrafiltration centrifuge tube (30 kD, Millipore) and centrifuged at 37 °C at 1500 g for 30 min (Thermo Fisher Scientific, USA) to obtain the filtrate. To evaluate the adsorption effect of the ultrafiltration membrane on PFAS, 700 μL of PFAS solution (2 μM, without protein) was processed under identical conditions. Each concentration was tested in triplicate. The filtrate and the incubated mixture were diluted 20- to 1000-fold with methanol, and following the addition of an internal standard, samples were stored at −20 °C for subsequent analysis. The quantification of PFAS was conducted using an ultra-high-performance liquid chromatography system (Waters, ACQUITY, USA) coupled with a triple quadrupole mass spectrometer (Waters, TQ-S, USA), as detailed in the SM Text S2 and Table S2.

2.1.4. Molecular docking prediction of PFAS to rL-FABP and RSA

Furthermore, the docking program AutoDock Vina (v1.1.2) was employed to predict the binding between 14 PFAS and both rL-FABP and RSA. The molecular structures of rL-FABP and RSA proteins were retrieved from the AlphaFold Protein Structure Database, while the 3D structures of PFAS compounds were acquired from PubChem. AutoDock Vina, AutoDock Tools 1.5.6, PyMOL(v2.4), and other tools were used to optimize the small molecules and proteins. Briefly, the AutoDock Tools were used to define the binding site boundaries for each protein structure individually. The output files were employed for docking with AutoDock Vina. The protein–ligand complexes were visualized and analyzed using PyMOL and Discovery Studio 2021. For RSA, six binding sites analogous to those in human serum albumin (HSA) were included, while rL-FABP encompassed the entire protein molecule. Redocking was performed using the crystal structures of HSA and perfluorooctane sulfonate (PFOS), following the methodology described by Ng and Hungerbuehler (2015). These redocking tests simulated the ability to reproduce results from experimental binding studies. For each compound, ten independent docking runs were executed, with the lowest energy conformation being chosen for further analysis.

2.2. PBPK modeling of PFAS in male rats

Following the previously published PBPK model framework (Cheng and Ng 2017; 2021), we developed a PBPK model for PFOS, perfluorohexane sulfonate (PFHxS) and perfluorooctanoic acid (PFOA). In contrast to the original model, our approach utilized independently obtained protein binding data as fixed parameters. The physiological compartments for rats included seven interconnected tissues: blood, liver, kidneys, intestines, muscle, fat, and the remaining body parts, all linked through blood circulation (Fig. 1) (Cheng and Ng 2017; 2021). Each tissue was divided into two sub-compartments: interstitial fluid and tissue. Following oral administration, PFAS were absorbed by the intestinal tissues, entered the bloodstream, and were subsequently transported to other tissues through cell membranes. Membrane transport was modeled as a combination of passive diffusion and transporter-facilitated transport, both of which were essential for cellular uptake. In the liver and kidneys, membrane protein transport is predominant, whereas passive diffusion is more significant in other tissues. The principal pathways of elimination and metabolism were through the kidneys and intestines, with renal elimination involving glomerular filtration, renal reabsorption, and renal clearance, while intestinal elimination mainly occured via fecal excretion.

Transporter-facilitated PFAS transport mechanisms were also incorporated into our PBPK model. Given the limited data on PFAS and transporter binding, we have included organic anion transporters (OATs), na/taurocholate co-transporting polypeptide (NTCP), and apical sodium-dependent bile salt transporter (ASBT) as representative transporters in our models. These transporters are essential for renal tubular excretion, reabsorption, and enterohepatic circulation (Weaver et al., 2010; Zhao et al., 2015; Zhao et al., 2017), and had been reported to have PFOA, PFOS, and PFHxS transport capabilities. The model was subsequently evaluated by comparing model predictions with multiple datasets from rat oral and IV experiments, with data sourced from the study by Kemper and Jepson (2003), Huang et al., (2019), Kim et al., (2016) and Kim et al., (2018). The model was implemented in R version 4.0, and parameter information are provided in the SM Table S3. Briefly, the physical parameters and partition parameters were obtained from previous studies (detailed in SM table S3), the protein binding data was available from this study, and other parameters were fitted using R package FME.

Regarding the model performance, we conducted a comparative analysis between the PBPK model for PFOA in male rats from prior studies and the root mean square error (RMSE) values derived from our current model. Pharmacokinetic parameters calculated from the model outputs were then systematically compared with experimentally observed reference values to assess whether the model's predictive performance aligned with empirical data. It should be noted that the primary objective of our PBPK model was to compare model performance with and without protein binding data. Therefore, additional validation datasets were not utilized.

2.3. Statistical analysis

Regression analyses and graphical representations were performed using R version 4.0 and Origin 2021, with $p < 0.05$ considered statistically significant.

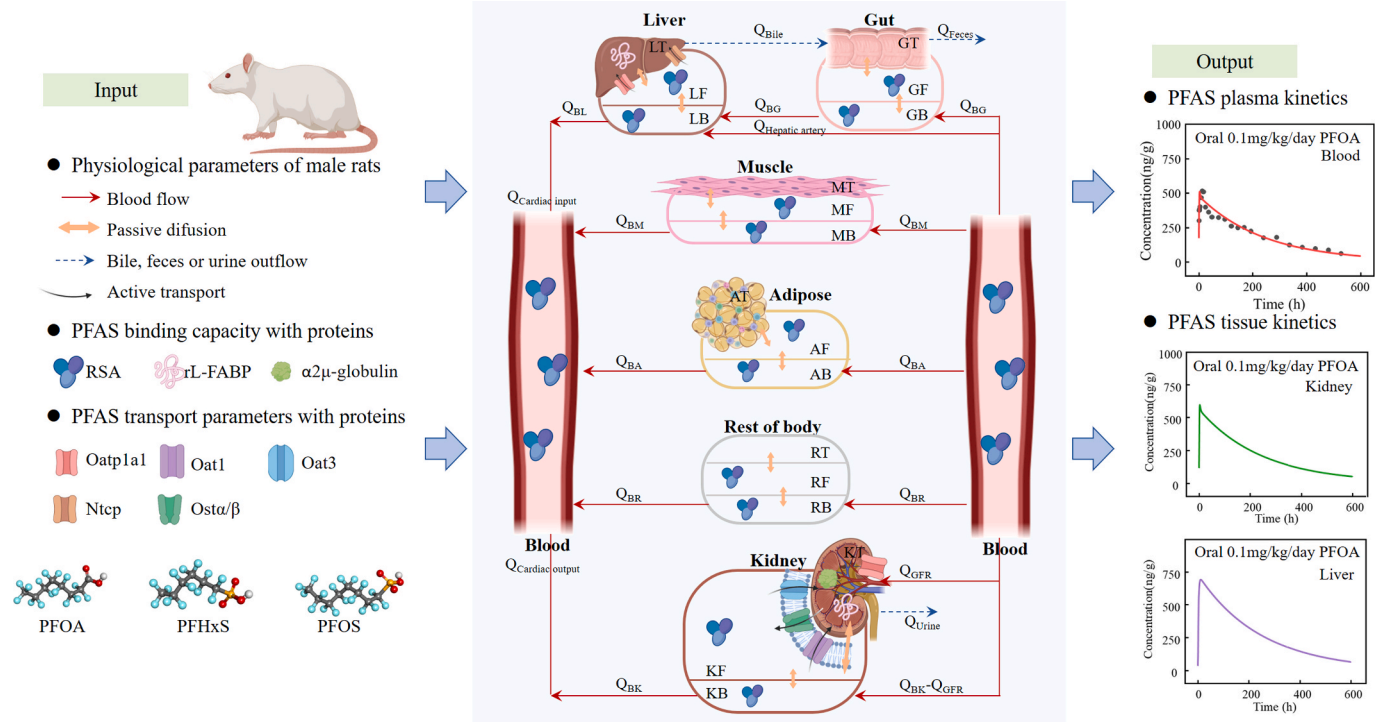


Fig. 1. The schematic representation of the rat PBPK model, includes seven tissue compartments: blood (B), liver (L), gut (G), muscle (M), adipose tissue (A), kidney (K), and the remaining body tissues (R). All compartments, except for blood, are divided into vascular (e.g., LB for liver) and tissue compartments. The tissue compartments are further subdivided into interstitial fluid (e.g., LF for liver interstitial fluid) and cellular tissue (e.g., LT for liver cellular tissue).

3. Results

3.1. Assessment of PFAS binding affinity with rL-FABP

This study has systematically investigated the binding affinities of 14 PFAS to L-FABP in rats (Table 1 and Fig. S3), with K_d values ranging from 5.36 to 132.16 μM (corresponds to K_a from 10^3 to 10^5 M^{-1}), revealing a single binding site. Among the PFAS tested, medium- and long-chain PFCAs (C > 6) exhibited higher binding affinities compared to short-chain PFCAs (C ≤ 6), with perfluorodecanoic acid (PFDA, C10) showing the strongest affinity (K_d = 5.36 μM). The binding constant of perfluorobutanoic acid (PFBA, C4) to rL-FABP was undetectable, possibly due to the fluorescence method's insensitivity or PFBA's low binding affinity. This finding is in agreement with the findings of Yang et al.(2020) on human L-FABP protein. Of the 3 PFAS regulated under the Stockholm Convention, PFOA displayed the highest affinity for rL-FABP (K_d = 12.30 μM), followed by PFHxS (K_d = 13.02 μM) and PFOS (K_d = 20.73 μM). Compared to PFSAs, PFCAs demonstrated a stronger affinity for rL-FABP, particularly for medium- and long-chain PFCAs. The dissociation constants (K_d) for the target PFAS binding to rL-FABP displayed a "U"-shaped trend relative to the number of carbon atoms (Table 1, Fig. 2a). Our results showed that the binding affinities of target short-chain PFAS to rL-FABP (K_d = 6.55 ~ 132.16 μM) were stronger than previously reported values (K_d = 6.07 ~ 608.40 μM) (Woodcroft et al. 2010).

Additionally, this study is the first to report the binding affinities of long-chain PFCAs, PFSAs, and GenX substances to the rL-FABP protein. On another aspect, the affinities of PFAS to rL-FABP were comparable to those measured for PFAS binding to human L-FABP using the same method (Zhang et al. 2013), though slightly lower than those obtained through equilibrium dialysis (Gao et al. 2019) (Table 1, Fig. 2a). Furthermore, the logarithm of the binding constants (K_a, $1/K_d$) for PFCAs tested in our study showed a strong correlation with both experimental membrane permeability (log $K_{mem/w}$, R^2 = 0.94, Fig. 2b) and predicted membrane permeability (R^2 = 0.72, Fig. 2c)) reported by Ebert et al. (2020).

3.2. The PFAS binding affinity with RSA

We employed ITC and ultrafiltration to evaluate the binding affinity and binding sites of PFAS to the macromolecular protein RSA. ITC analysis showed that the K_d of the tested PFAS ranged from 31.18 to 303.08 μM (corresponds to K_a from 10^4 to 10^5 M^{-1}), with approximately 1–4 binding sites. Among these, PFHxS exhibited the strongest binding with RSA (K_d = 26.87 μM, n = 2.15), and PFOA (K_d = 26.77 μM, n = 3.11) and its alternative GenX (K_d = 17.87 μM, n = 2.47) demonstrated slightly lower binding affinities compared to PFHxS (Fig. S4). The affinity of PFCAs for RSA was slightly higher than that of PFSAs, with medium- and long-chain PFCAs showing stronger affinity compared to short-chain PFAS, consistent with results observed for rL-FABP (Table 1).

Compared to the ITC approach, ultrafiltration indicated relatively higher binding affinities and a greater number of binding sites for PFAS and RSA, with affinities around 10^4 M^{-1} and 3–5 binding sites. Notably, PFBS exhibited particularly strong affinity for RSA, with a K_d of approximately 2.78 μM (Table 1). The only reported measurement of PFOA's binding affinity to RSA (K_d = 360 μM, n = 7.8) (Han et al. 2003) is lower than the values obtained in this study (K_d = 26.77 μM, n = 3.11), but it indicates a greater number of binding sites. Our study demonstrated that the binding affinity of PFAS to RSA was stronger than that to HSA, as previously reported by Gao et al. (2019) based on human protein, which identified a single binding site using equilibrium dialysis.

The binding affinities and site numbers determined by ITC were lower than those derived from ultrafiltration, likely due to PFAS adsorption on the regenerated cellulose membrane within the ultrafiltration tubes (Liu et al., 2019). Overall, most of the binding affinities of tested short-chain PFAS to RSA yielded a good fit between using the ultrafiltration and ITC method, while long-chain PFCAs showed opposite appearance (Fig. S5). The reason for the poor fits for long-chain PFCAs could be explained by excessive adsorption. Our results revealed that the

Table 1
The summary on measured and reported dissociation constants (K_d, μM) of PFAS binding to FABP and serum proteins in different species.

	rL-FABP		hL-FABP		RSA						HSA	BSA
	$K_{d\text{-}FABP}$[a]	$K_{d\text{-}FABP}$[b]	$K_{d\text{-}FABP}$[c]	$K_{d\text{-}FABP}$[d]	Ultrafiltration Method		ITC		n[e]	$K_{d\text{-}RSA}$[e]	$K_{d\text{-}HSA}$[d]	$K_{d\text{-}BSA}$[f]
					n	$K_{d\text{-}RSA}$	n	$K_{d\text{-}RSA}$				
	(this study)	(previous studies)	(previous studies)		(this study)				(previous studies)		(previous studies)	
Perfluorocarboxylic acids												
PFBA	–-		ND	879 ± 51	4.46 ± 0.21	16.21 ± 2.95	2.84 ± 0.35	168.40 ± 33.08			419 ± 37	
PFPeA	27.57 ± 3.68	608.40 ± 16.06	ND	685 ± 23	3.92 ± 0.16	13.9 ± 1.86	3.33 ± 0.21	303.08 ± 111.38			286 ± 17	
PFHxA	132.16 ± 6.16	261.70 ± 7.00	ND	443 ± 25	3.36 ± 0.14	31.36 ± 2.86	1.93 ± 0.65	166.70 ± 33.24			195 ± 14	
PFHpA	44.54 ± 1.12	61.17 ± 1.95	336.0 ± 40.3	358 ± 16	3.02 ± 0.62	10.15 ± 1.37	2.79 ± 0.15	79.96 ± 8.83			172 ± 23	
PFOA	12.30 ± 3.60	13.14 ± 0.27 23.81 ± 0.89 (n = 2)	50.4 ± 4.0	197 ± 13	4.24 ± 0.52	26.77 ± 2.77	3.11 ± 0.33	94.54 ± 8.16	7.8 ± 1.5	360 ± 80	115 ± 16	2.65
PFNA	6.55 ± 0.19	6.07 ± 2.70 9.84 ± 3.95 (n = 2)	16.2 ± 0.6	90 ± 8	4.21 ± 0.49	13.27 ± 1.20	1.06 ± 0.05	31.18 ± 2.45			73 ± 21	0.23
PFDA	5.36 ± 0.73		12.9 ± 0.6	77 ± 14	3.70 ± 0.32	10.33 ± 0.85	1.91 ± 0.49	161.20 ± 9.41			65 ± 8	0.16
PFUnDA	6.95 ± 0.19		10.6 ± 1.2	54 ± 5			1.42 ± 0.3	59.18 ± 9.79			79 ± 13	
PFDoDA	13.46 ± 4.18		12.3 ± 0.7	95 ± 11			1.17 ± 0.95	32.78 ± 21.41			103 ± 12	
PFTeDA	6.32 ± 1.09		60.5 ± 6.7	317 ± 20			1.02 ± 0.09	195.30 ± 8.54			265 ± 21	
Perfluorosulfonic acid												
PFBS	110.82 ± 4.80		1034 ± 110	436 ± 33	3.68 ± 0.14	2.78 ± 0.58	1.59 ± 0.59	176.30 ± 39.11			169 ± 15	10.75
PFHxS	13.02 ± 0.11		85.7 ± 11.8	210 ± 14	4.24 ± 0.53	78.87 ± 2.74	2.15 ± 0.18	26.87 ± 1.18			79 ± 12	1.54
PFOS	20.73 ± 0.90		18.5 ± 0.6	81 ± 7	4.61 ± 0.32	13.77 ± 2.65	2.79 ± 0.40	129.80 ± 17.54			38 ± 5	0.56
Perfluorinated alternatives												
GenX	108.23 ± 6.47		–-	–-	3.26 ± 0.23	17.87 ± 5.2	2.47 ± 0.4	67.57 ± 20.34				15.38

[a] In this study, K_d for PFAS binding to rL-FABP expressed in prokaryotic cells was determined using the 1.8-ANS fluorescence displacement method.
[b] Woodcroft et al.(Woodcroft et al. 2010) measured the K_d for PFAS binding to rL-FABP expressed in Escherichia coli (E. coli) using the 1.8-ANS fluorescence displacement method.
[c] Zhang et al.(Zhang et al. 2013) determined the K_d for PFAS binding to hL-FABP expressed in E. coli using the nitrobenzoxadiazole-labeled lauric acid (NBD-lauric acid) fluorescence displacement method.
[d] Gao et al,(Gao et al. 2019) calculated the K_d for PFAS binding to hL-FABP expressed in E. coli and commercially sourced human serum albumin using equilibrium dialysis.
[e] Han et al.(Han et al. 2003) determined the K_d for PFAS binding to rat plasma proteins using size-exclusion chromatography (SEC) and ^{19}F nuclear magnetic resonance (NMR).
[f] Alesio et al.(Alesio et al. 2022) calculated the K_d for PFAS binding to bovine serum albumin (BSA) using equilibrium dialysis.

ultrafiltration tubes can adsorb 20–95 % of PFAS at lower concentrations (2 μM), with adsorption loss being significantly correlated with log K_{ow} (as predicted by *EPI*) ($p < 0.001$) (Fig. S5). Hence, considering the removed membrane adsorption interference and lower background noise associated with the ITC method, ITC-derived results were used in subsequent analysis.

3.3. Molecular docking prediction of PFAS binding to proteins

To elucidate the binding conformation and binding energy (ΔG) between the tested PFAS and both rL-FABP & RSA, molecular docking simulations were further employed to model the ligand–protein complexes. The results demonstrated that all tested PFAS spontaneously bind to both rL-FABP and RSA (ΔG < 0), indicating favorable exothermic processes, potentially driven by increased entropy due to desolvation of the hydrophobic fluorinated tails. Notably, several long-chain PFAS (C > 11) adopt a "U"-shaped conformation within the ligand-binding cavities of both rL-FABP and RSA (Figs. 3, 4, S5–S6, and S9–S11).

All tested PFAS were observed to bind within the fatty acid-binding cavity of rL-FABP, with some also interacting with a secondary hydrophobic site through hydrophobic interactions. For medium- and short-chain PFCAs (chain length < 8), the charged head groups interact with methionine (Met), valine (Val), and threonine (Thr) residues upon entering the rL-FABP cavity, while the hydrophobic tails contact nonpolar residues within the cavity (Table S4). For short-chain PFCAs, the carboxylate tail fully extends into the cavity, maximizing hydrophobic interactions with surrounding amino acid residues. However, when the carbon chain exceeds nine, the carboxylate tail must bend to fit within the cavity, and these longer chains do not enter the secondary hydrophobic binding region. For PFCAs with chain lengths > 9, the charged head group and the mid- and tail-sections of the molecule form hydrogen bonds with amino acid residues, leading to a U-shaped conformation as it enters the cavity (Figs. 3, 4, and S6, S7).

Molecular docking results demonstrated that PFHpA had the highest number of binding pockets with RSA, identifying six distinct docking

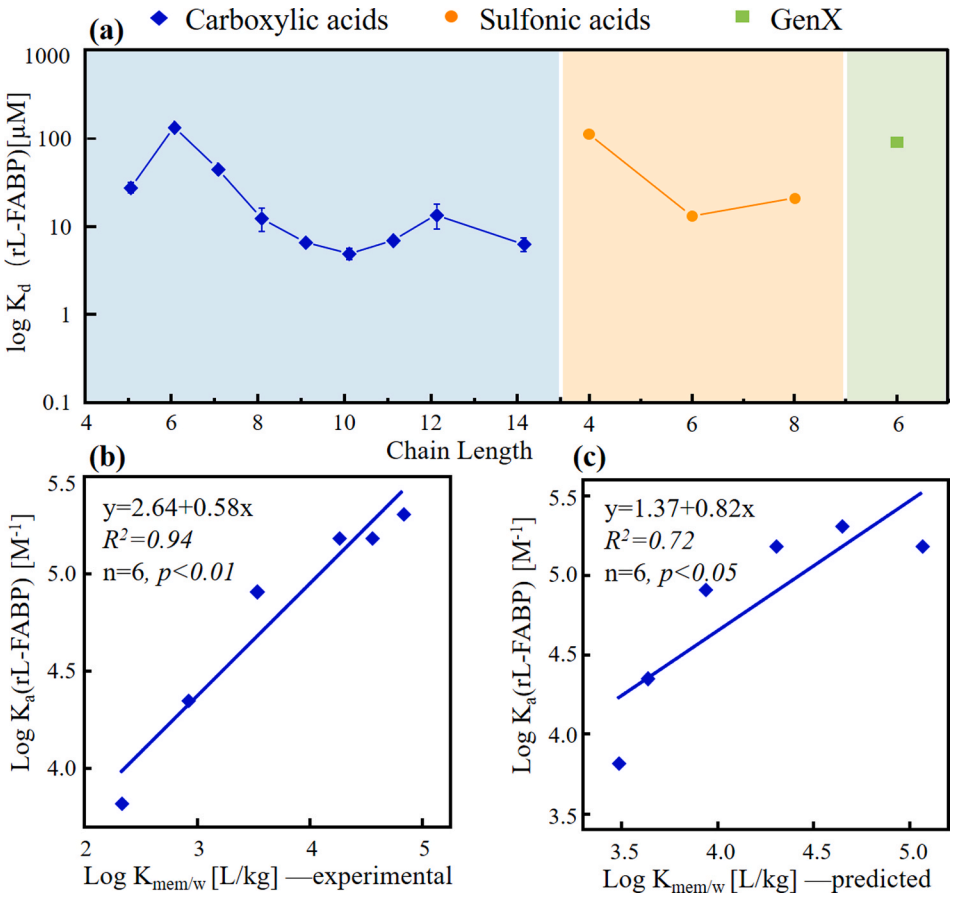


Fig. 2. The binding capacity of tested PFAS to rL-FABP: (a) the relationship between PFAS binding affinity to rL-FABP and carbon chain length; (b) the correlation between PFCAs binding affinity to rL-FABP and the experimentally determined membrane/water partition coefficient ($K_{mem/w}$); and (c) the relationship between PFCAs binding affinity to rL-FABP and the predicted membrane/water partition coefficient ($K_{mem/w}$).

sites (Fig. S9). Conversely, PFCAs with C > 11 only bind to RSA within the cleft between IIB and IIA in a bent U-shaped conformation, in reference to the conformation of HSA (Fig. S8, S9). Among the targeted PFAS, ten compounds bind in this region, with their polar head groups (carboxyl or sulfonic acids) forming stable hydrogen bonds and polar interactions with ionizable residues, thereby achieving stability. Leucine (Leu), lysine (Lys), and glycine (Gly) play critical roles, as at least one of these residues is present in the PFAS binding pockets within the cleft between IIB and IIA (Table S5). The polarizability of organic fluorine enables participation in short hydrogen bond interactions(Hathwar et al., 2014), facilitating polar interactions between carbon–fluorine tails and the side chains and backbones of RSA residues (Maso et al., 2021). Similar to the binding behavior observed with HSA and GenX (Crisalli et al., 2023), the ether oxygen in GenX is not involved in any significant side-chain interactions that would notably enhance binding (Fig. S11).

3.4. PBPK modeling of PFOA, PFOS and PFHxS

Based on the protein data obtained in 3.1 and 3.2, we have established the PBPK model for PFOA, PFOS and PFHxS, by incorporating PFAS with protein binding and transporter data (Fig. 5), and effectively predicted the concentration of PFAS. Notably, the model demonstrated a generally superior fit for PFOA when independent protein binding parameters reported in our study were used, particularly in the case of oral exposure at 1 mg/kg/day, where RMSE decreased by approximately 38 %, reducing from 1184.10 to 741.60 (Fig. S12). Specifically, our model closely aligns with experimental values for the maximum plasma concentration (C_{max}), half-life ($T_{1/2}$), and clearance (CL) of PFOA (Table S6), with the prediction error reduced by 20 %~216 %. The clearance is approximately 24.54 mL/kg/day, aligning closely with the experimental value of 20.9 ± 3.79 mL/kg/day and providing a more accurate estimate than the original PBPK model's prediction of 69.64 mL/kg/day. Our modeling results show that the shortest half-life for oral exposure to PFOA at 1 mg/kg/day is approximately 4.67 days, closer to the experimental value of 5.76 ± 1.33 days. Predicted half-lives for oral exposure to PFOA at 0.1 mg/kg/day and intravenous exposure at 1 mg/kg/day are 7.04 and 7.81 days, respectively, which is also close to the experimental values of 8.41 ± 1.56 days and 7.73 ± 0.82 days, respectively.

The model also achieves sound fits for PFOS and PFHxS at low to moderate concentrations regarding C_{max}, T_{max} (time to maximum concentration), $T_{1/2}$, and CL. Nonetheless, at high concentrations, the model overestimates the clearance of PFHxS by approximately 3.5-fold compared to the experimental results, potentially due to variations in protein uptake rates or binding capacities under high exposure conditions (Table S6). Overall, in contrast to previous models lacking binding data, this study quantified the binding affinities and their substitutes to RSA and rL-FABP proteins, triggering a better fitness on the toxicokinetics of PFAS.

It should be noted that the primary objective of our PBPK model was to compare model performance with and without protein binding data. Therefore, additional validation datasets were not utilized. Also, numerous existing studies have constructed PBPK models without independent validation due to limited availability of experimental animal data. For instance, the works of Chen et al., (2024), Zhu et al., (2023a) and Zhu et al., (2025) developed PBPK models based solely on their own experimental results, where model predictions were deemed compliant

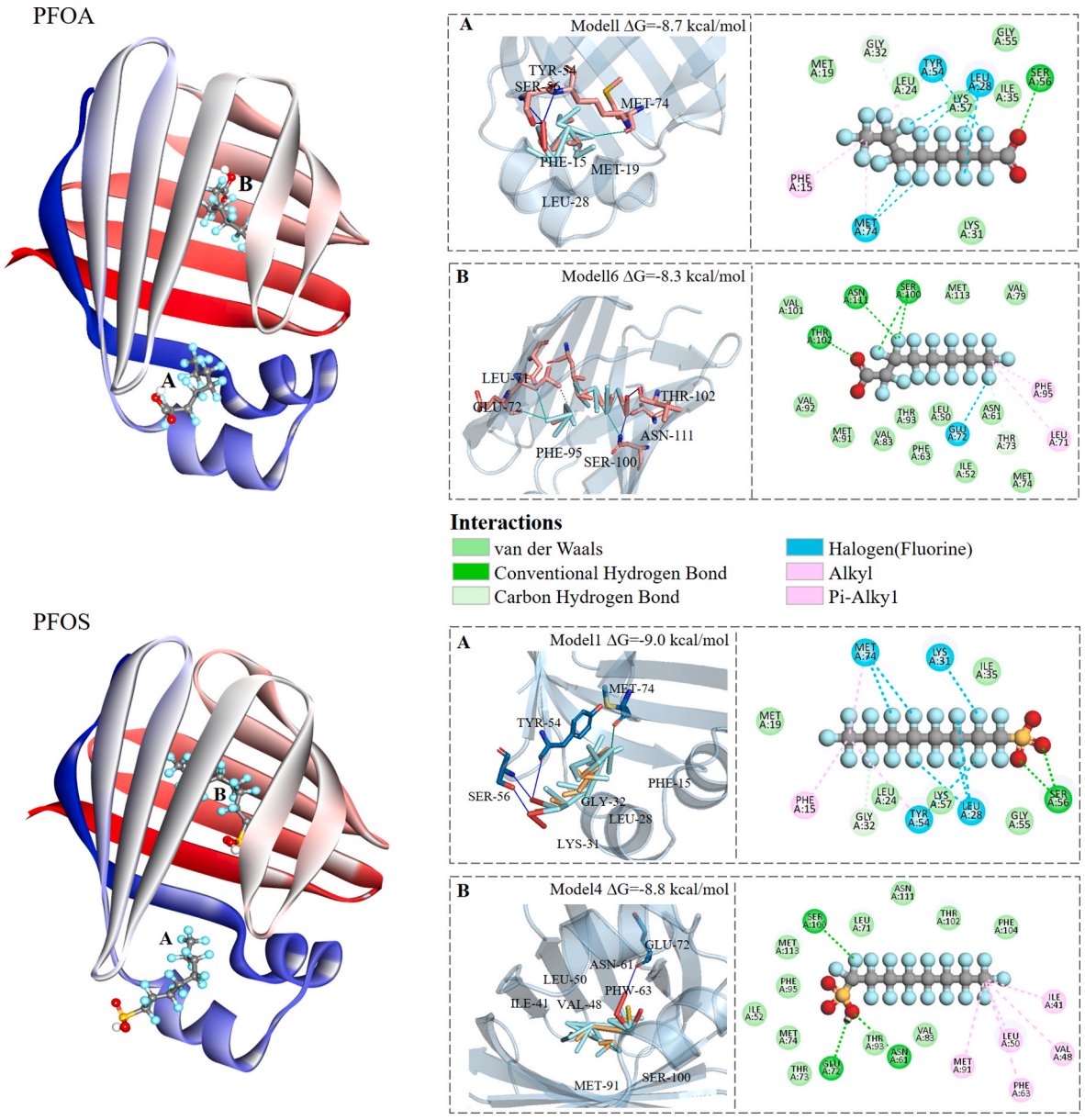


Fig. 3. Conformational analysis of PFOA and PFOS binding to rL-FABP.

with the WHO-recommended model accuracy criteria (i.e., predictions within a two-fold margin of error relative to measured values).

4. Discussion

To the best of our knowledge, this study represents the first comprehensive investigation of the binding affinities ($K_a = 10^3 \sim 10^5$ M^{-1}) of 14 PFAS and their substitutes to rL-FABP and RSA. The results indicated that most tested PFAS exhibited slightly higher affinities for rL-FABP compared to RSA, while RSA possesses a greater number of binding sites. Molecular docking was utilized to confirm the binding of PFAS with these proteins, identifying specific binding sites, key amino acid residues, and involved molecular interactions. Moreover, experimentally determined protein binding and transport parameters were utilized to construct permeability-limited PBPK models for male rats, accurately predicting the toxicokinetics of PFAS in blood and tissues. It is noteworthy that our experimental and molecular docking analyses specifically quantified the binding constants of individual PFAS-protein. However, under physiological conditions where protein concentrations in biological systems vastly exceed multiple PFAS exposure levels (serum PFAS: 0.1–50 nM vs. albumin: 500–700 μM) (Wallis et al., 2024), saturation of protein binding sites does not occur. Consequently, competitive binding between different PFAS congeners was not incorporated into our current framework, as their molar ratios (<0.01 %) render inter-PFAS competition negligible compared to dominant endogenous ligand displacement effects.

4.1. Measurement and computational approaches for PFAS-Protein binding

In this study, when measuring the binding affinity between the macromolecular protein RSA and PFAS, the ITC method demonstrated reduced background interference compared to the ultrafiltration method, since the ultrafiltration method may be influenced by non-specific adsorption of PFAS to the regenerated cellulose membrane. The selection of appropriate experimental methods for measuring PFAS-protein binding requires careful consideration of both the protein and chemical properties. Methods are generally classified into indirect and

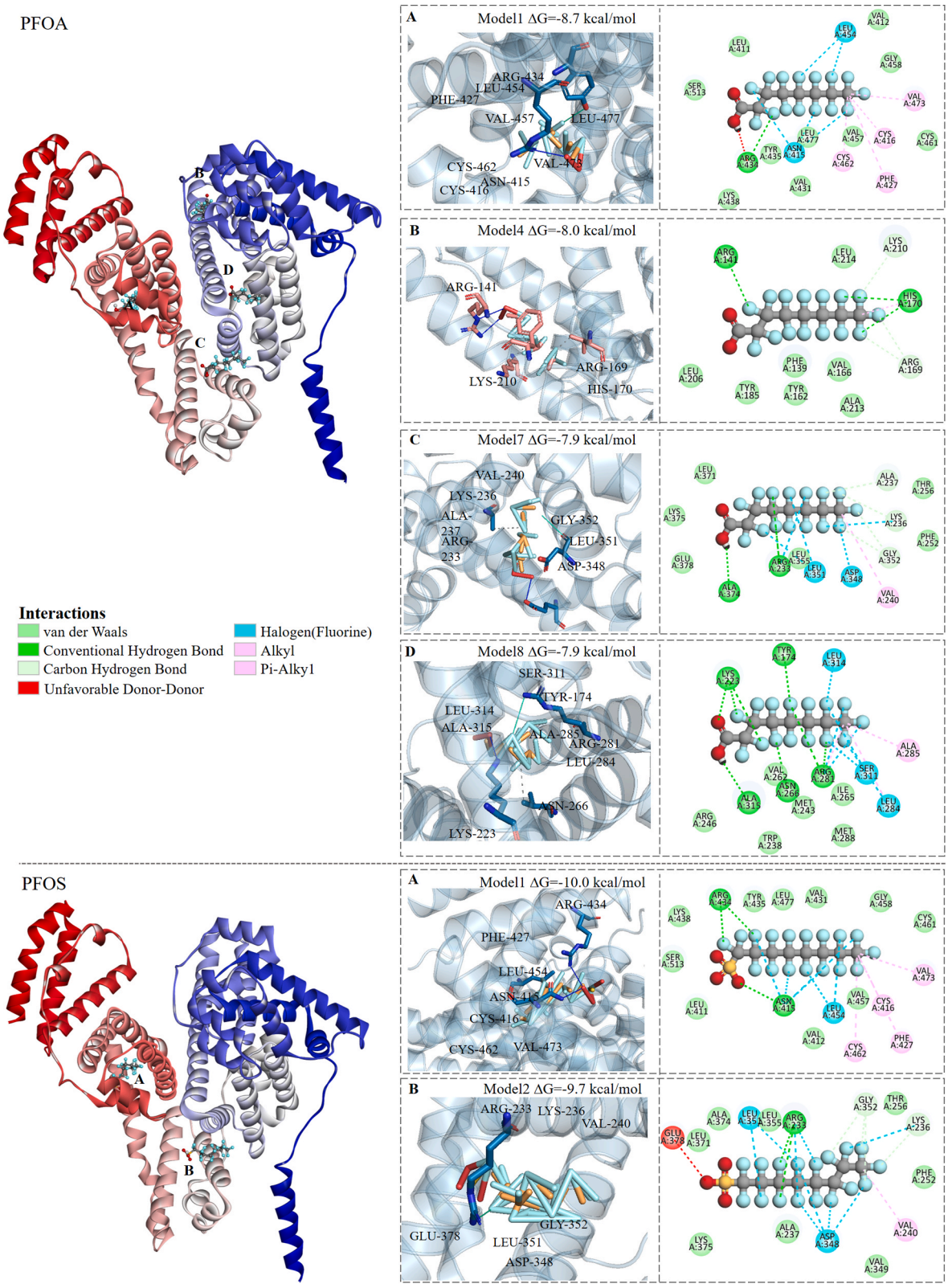


Fig. 4. Conformational Analysis of PFOA and PFOS binding to RSA.

direct approaches. For small proteins such as L-FABP, which are known to have only 1–2 binding sites and produce specific fluorescence effects when interacting with fluorescent probes, fluorescence-based methods can efficiently measure the binding capacity and number of binding sites (Gao et al. 2019; Woodcroft et al. 2010; Zhang et al. 2013). However, for large proteins such as RSA, which possess 4–9 potential drug-binding sites, direct methods for determining binding constants are preferred (Mishra and Heath 2021; Sheng et al., 2020; Tiburtini et al., 2024).

The ITC method provided comprehensive thermodynamic data, including ΔH, K_a, and n. It requires both the protein and small molecule ligand to be dissolved in identical solutions, with strict requirements for ligand solubility and concentration ratios to minimize the impact of

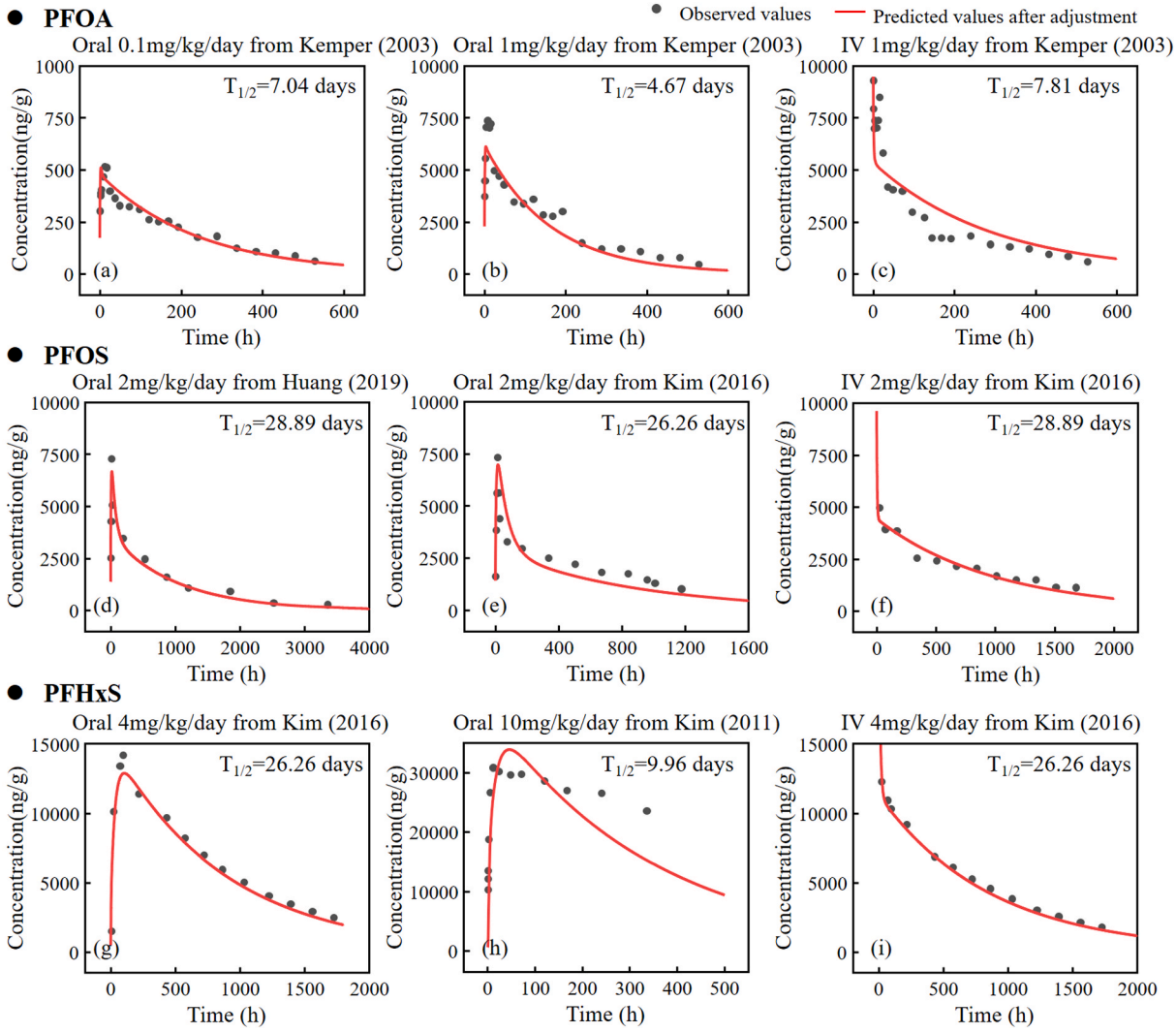


Fig. 5. The pharmacokinetics of PFAS in blood and half-life predictions across varying exposure concentrations: (a), (b) & (c): blood concentration–time profiles of PFOA, with data sourced from Kemper et al. (2003)(Kemper and Jepson 2003); (d): blood concentration–time profile of PFOS, using data from Huang et al. (2019) (Huang et al. 2019), (e) & (f) blood concentration–time profile of PFOS, utilizing data from Kim et al. (2016)(Kim et al. 2016); (g) & (i): blood concentration–time profile of PFHxS sourced from Kim et al. (2016)(Kim et al. 2016), (h): PFHxS data from Kim et al. (2018)(Kim et al. 2018). The gray points represent experimental measurements, while the red lines represent the fitted values from this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

background dilution heat. Ultrafiltration, which rapidly separates small molecule ligands from complexes using a semipermeable membrane, encounters challenges due to the substantial adsorption of PFAS ligands onto the regenerated cellulose membrane. This adsorption mediates affects the formation of PFAS-protein complexes and is related to the ligand's K_{ow} (Fig. S5). For long-chain PFAS, the adsorption rate on the ultrafiltration membrane can exceed 90 % under some scenarios. Similarly, although equilibrium dialysis is frequently referred to as the "gold standard," it is also influenced by the regenerated cellulose membrane, making it suboptimal for measuring PFAS-protein binding. Also, some other methods, surface plasmon resonance (SPR), X-ray crystallography, and nuclear magnetic resonance (NMR) enable precise determination of binding constants (Ka) and stoichiometric parameters (n), yet remain constrained by technical complexity in protein immobilization (SPR), substantial sample requirements (NMR), and prolonged experimental timelines (X-ray crystallography). UV–Vis spectroscopy, microscale thermophoresis (MST), and dynamic light scattering (DLS) exhibit compromised measurement accuracy due to inherent methodological limitations – environmental sensitivity in MST, particle polydispersity interference in DLS, and spectral overlap artifacts in UV–Vis analyses. In Table S7, we have summarized the advantages and disadvantages of these measurement approaches. Discrepancies may arise when determining the binding affinity between proteins and PFAS using different methodologies, which can be attributed to systematic differences among the methodologies (Alesio et al., 2022; Li et al., 2021; Peng et al., 2024).

AutoDock Vina is a widely recognized computational tool for predicting the conformations and binding energies of PFAS and proteins, providing detailed insights into the key binding forces and critical amino acid residues. In our study, discrepancies between the predictions and experimental data were observed, suggesting that molecular docking alone may not fully capture the complexities of PFAS-protein binding. For most PFAS-RSA docking results, the number of binding sites determined by ITC experiments correlated with those predicted by molecular docking ($p < 0.05$); however, discrepancies were also observed. Theoretically, the binding affinity, K_a, can be estimated using binding (Ng and Hungerbuehler 2015) energy ΔG ($K_a = e^{-\Delta G/RT}$), while our findings demonstrate discrepancies between the K_a values predicted by molecular docking and those determined experimentally (Table S8). These inconsistencies may arise from the influence of pH, buffer composition, and solubility at different concentrations on PFAS-protein binding under

physiologically relevant conditions.

4.2. Interspecies variations in PFAS binding affinities to L-FABP and serum albumin

Understanding the binding affinities of proteins to PFAS across different species is essential for elucidating the kinetic disparities between animals and humans. Although rL-FABP and RSA exhibit structural similarities with human L-FABP (hL-FABP) and HSA, differences in amino acid sequences, structural conformations, and ligand-binding characteristics result in variations in PFAS binding affinities. We compared the binding affinities of tested PFAS for rat proteins in this study to those reported for humans and found that most tested PFAS exhibited higher binding affinities to rat proteins compared to human proteins (Table 1), particularly for L-FABP. Among them, perfluorotetradecanoic acid (PFTeDA) demonstrated the most significant difference, exhibiting a binding affinity to rL-FABP that was 9.6 times stronger than to hL-FABP (Zhang et al. 2013). Due to the structural similarities between these proteins, the dissociation constant (K_d) values for PFAS binding to rL-FABP obtained in our study showed a strong correlation with those measured for hL-FABP using fluorescence (R^2 = 0.97, $p < 0.001$) or equilibrium dialysis ($R^2 = 0.55$, $p < 0.1$).

4.3. The association between PFAS Liver-Blood partition coefficients and protein binding

To investigate the liver-blood partition and PFAS binding to proteins, we collected liver-blood partition coefficients for both rats and humans (Table S9). In male rats, the liver-blood distribution ratios for PFHxA (Dzierlenga et al., 2020), PFBS (Huang et al. 2019), and PFHxS (Olsen et al., 2009) were less than 1, whereas those for PFOA (Dzierlenga et al. 2020), PFDA (Dzierlenga et al. 2020), PFOS (Kim et al. 2016), and GenX (Huang et al. 2019) exceeded 1, with PFDA presenting the highest ratio, approximately 5.86. In female rats, only PFDA (Dzierlenga et al. 2020) and PFOS (Kim et al. 2016) exhibited liver-blood distribution ratios greater than 1, with PFHxS (Olsen et al. 2009) showing the lowest ratio, around 0.24. This study gathered the liver-blood partitioning ($R_{L/P}$) of PFAS in rats and identified a positive correlation with the K_a (rL-FABP)/K_a (RSA) ratio (Fig. 6(a)&(b)), suggesting a competitive distribution mechanism of PFAS between the liver and blood. Notably, the correlation between $R_{L/P}$ and K_a (rL-FABP)/K_a (RSA) was weaker in female rats ($R^2 = 0.67$) compared to male rats ($R^2 = 0.93$). This discrepancy may be attributed to sex-based differences in liver-blood partition coefficients in rats, potentially influenced by sex-dependent markers in the liver and kidneys following PFAS exposure (Roth et al., 2021). The steeper slope of the linear correlation observed in male rats compared to female rats may be caused by elimination variations.

Similar observations were noted when the association was extended to human. A positive correlation was also found when analyzing the liver-blood partition coefficients in humans ($R_{L/P}$, without gender distinction) in relation to the binding affinities of human proteins (K_a (hL-FABP)/K_a (HSA), Table 1) for PFAS (R^2 = 0.67, Fig. 6(c)). This finding confirms that the high affinity of PFAS for L-FABP and serum albumin results in elevated concentrations in both the liver and blood, with PFAS competitively binding to both. This finding provides an effective and feasible method for estimating the concentration of PFAS in human liver. For instance, the known blood PFHxS concentration in the Danish population is 0.153 ng/g, the ratio K_a (hL-FABP)/K_a (HSA) = 0.38, corresponding to an estimated liver concentration of approximately 0.084 ng/g, which is close to forensic autopsy results of 0.074 ± 0.052 ng/g (Nielsen et al. 2024).

Jia et al. (2022) reported that the liver-blood partition coefficients in mice, along with the population liver-blood partition coefficients ($R_{L/S}$) estimated from the liver-blood volume ratios of mice and humans, are negatively correlated with the binding affinities of human L-FABP and HSA (log K_d (hL-FABP μM)/log K_d (HSA μM)). The positive rather than negative association found in our study can be attributed to unit conversions. Moreover, our findings extend previous research to rats, and a recent study suggest that rat liver (comparing to mouse) serves as a more suitable model for predicting PFAS distribution in human tissues than mouse liver (Ryu et al. 2024).

5. Conclusion

In conclusion, this study measured the binding affinities of 14 PFAS compounds to rL-FABP and RSA. These values are essential for understanding PFAS bioaccumulation and serve as key inputs for modeling studies. Utilizing these binding affinities, PBPK model were developed for various PFAS, achieving precise fitness of their toxicokinetics. The results also reveal that competitive binding of certain PFAS to L-FABP and serum albumin partially drivers their partition between the liver and blood in rats, as well as in humans. This study establishes mechanistic linkages between PFAS-protein binding dynamics and toxicokinetic models, enabling cross-species organ-specific concentration predictions. By proposing a QIVIVE (Quantitative In Vitro-to-In Vivo Extrapolation) workflow that translates binding affinities into tissue-specific distribution coefficients, we bridge molecular interaction data with organism-level toxicokinetic profiling. These advances provide critical insights for future investigations into PFAS toxicokinetic mechanisms, offering a predictive framework to refine chemical prioritization strategies in ecological and human health risk assessments.

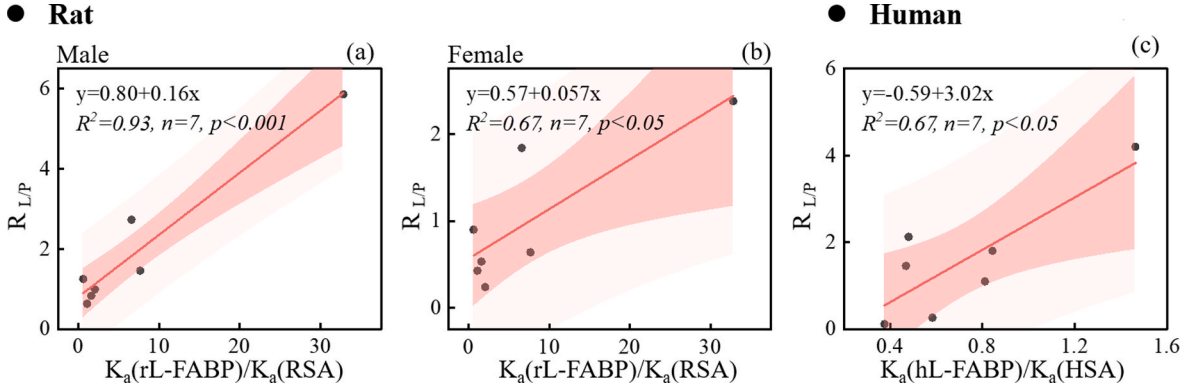


Fig. 6. The association between binding affinity ratio FABP: RSA proteins and the liver-plasma partition coefficient. (a): male rats (b): female rats $R_{L/P;}$ (c): humans; (RSA) with human $R_{L/P}$ for PFAS(Baumert et al. 2023). The red shaded area indicates the 95% confidence and prediction intervals. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

CRediT authorship contribution statement

Xiarui Fan: Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation. **Xiaomin Li:** Methodology, Investigation. **Tong Li:** Resources, Methodology. **Bing Shao:** Resources, Methodology. **Shan Niu:** Writing – review & editing, Methodology, Investigation. **Wenhong Fan:** Resources, Methodology. **Qi Wang:** Methodology, Investigation. **Lv Zhong:** Methodology. **Xiangrui Wang:** Methodology, Investigation. **Ziwei Wang:** Investigation, Data curation. **Fujun Ma:** Methodology, Investigation, Formal analysis. **Guomao Zheng:** Supervision, Methodology. **Hui Peng:** Writing – review & editing, Supervision, Methodology, Investigation. **Lili Chen:** Software, Conceptualization. **Zhaomin Dong:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2025.109591.

Data availability

Data will be made available on request.

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